

DSS Project Proposal for DIGITALISATION – for Port Traffic Management

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Problem Description:

System's Goal to Achieve:

The goal is to optimise pre-gate parking management at small EU ports, enhancing operations for varying traffic types and ensuring efficient use of space and resources. Key considerations include service traffic coordination and safety protocols.

Criteria for Decision Support System (DSS):

1. Budget constraints for parking layout changes.
2. Maximization of the number of service areas.
3. Minimization of time to find a parking slot.
4. Optimized traffic flow to reduce queuing times.
5. Safety measures for COVID-19.
6. Accommodation for peak and off-peak seasons.
7. Compliance with environmental regulations.
8. Utilization rates of truck and car parking slots.
9. Flexibility for future expansion or contraction of spaces.
10. Efficiency of check-in gate operations.
11. Contingency plans for emergencies or incidents.
12. Maintenance and operational costs of parking facilities.
13. Adaptability to different vehicle sizes and types.
14. Impact on local traffic outside the port area.

15. User experience and satisfaction levels.
16. Security measures against unauthorised access.
17. Integration with existing port infrastructure.
18. Digital ticketing and payment systems.
19. Data collection for continuous improvement.
20. Scalability to other ports with similar challenges.

General Architecture of DSS:

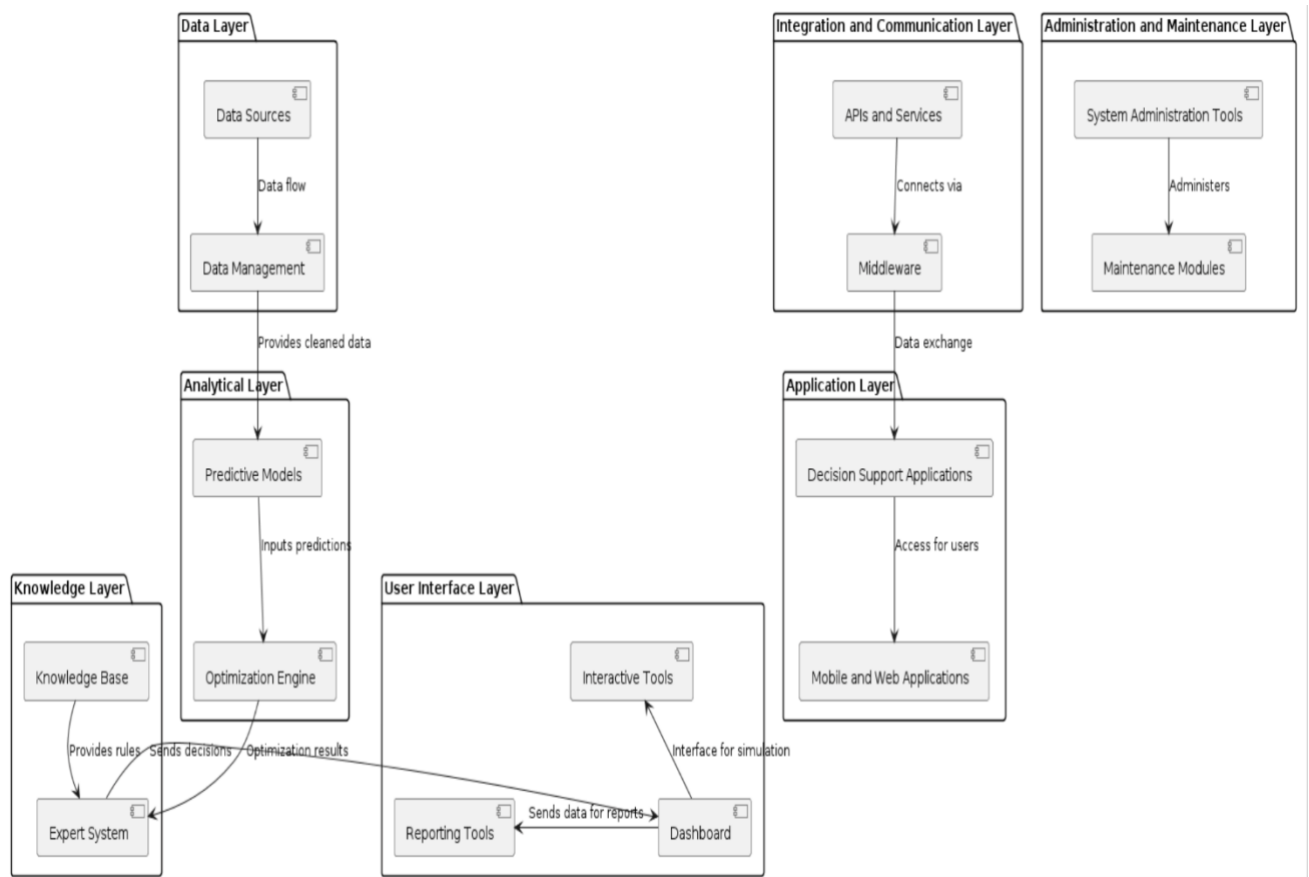


Diagram: the general architecture of Intelligent DSS.

Database Description:

The database serves as the backbone for the Intelligent DSS, holding all the pertinent data necessary for making informed decisions.

- **Traffic Data:** Records of vehicle entry and exit times, types, and durations of stay.
- **Parking Data:** Layouts of parking areas, current occupancy, and historical utilization trends.
- **Operational Data:** Ferry schedules, ticketing information, and gate operations.

This data should be structured to support quick retrieval and analysis, likely in a relational database that can handle complex queries and support concurrent operations.

Model Description:

The modelling aspect of the DSS will be responsible for processing inputs and generating predictions and recommendations.

- **Queuing Theory Models:** These models will simulate traffic flow, predicting bottlenecks and optimal throughput of vehicles.
- **Layout Optimization Models:** Algorithms designed to allocate parking spaces dynamically, maximizing usage while minimising congestion.
- **Predictive Analysis Models:** Machine learning algorithms that can forecast peak times and suggest proactive measures.

Knowledge Base Description:

- **Rule-Based Systems:** Contain rules for assigning parking based on vehicle type, size, and expected duration of stay.
- **Expert System:** Encapsulates expert knowledge of port operations to guide the decision-making process in complex scenarios.

Explanation of the Implementation:

Tools:

- **Simulation Software:** Used for creating and testing different traffic and parking scenarios in a virtual environment.
- **Database Management System (DBMS):** Software for managing databases, ensuring data integrity, security, and availability.

Techniques:

- **Machine Learning (ML):** Algorithms will analyse data trends to make predictions about parking demand and traffic flow.
- **Heuristics:** These are problem-solving approaches that quickly yield good enough solutions where finding an optimal solution is impractical due to complexity.
- **Data Analytics:** Involves analysing raw data to make conclusions about information, helping to understand patterns and trends for decision-making.